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Enhancing Well Integrity Across Weak Formations Using Foamed Cement Systems: A Detailed Analysis

S. Alharbi, T. Heinold, D. Bedford, F. Alqahtani, S. Husaini, and A. Almuraisel, Saudi Aramco, Dhahran, Kingdom of Saudi Arabia; Z. Bouhenaf, Z. Bechah, and A. Elhancha, SLB, Dhahran, Kingdom of Saudi Arabia

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Abstract

The paper explores the challenges of cementing wellbores in high permeable and fragile formations, traditional differential valve (DV) stage tools often face issues such as mechanical failures, cement channeling, and high costs. As a promising alternative, foamed cement offers enhanced solutions due to its lightweight nature, high compressive strength, low permeability, and durability. The process involves meticulous design and material selection, using specialized equipment to incorporate gaseous nitrogen for uniform distribution in the slurry. Pre-job planning, real-time monitoring during execution, and post-job evaluations are critical to ensuring quality and effectiveness, ultimately achieving reliable wellbore integrity and minimizing risks.

Foamed cement excels over DV stage tools. It provides superior zonal isolation and formation stability. It prevents fluid migration and maintains well integrity. Foamed cement is robust under high-temperature and high-pressure conditions. It minimizes the risk of fracturing fragile formations. Foamed cement reduces operational delays. It enhances job success rates. It offers cost savings by decreasing remedial operations. Comparative analysis highlights foamed cement's advantages. It overcomes challenges in lifting the cement column. It avoids extra interventions. The paper underscores foamed cement's innovative aspects. It emphasizes unique properties and advanced formulations. Foamed cement has successful operational applications. It demonstrates improved zonal isolation and well integrity. Real-time monitoring and post-job evaluations establish foamed cement as a superior solution. It addresses traditional issues associated with DV stage tools.

Background

The oil and gas production industry encounters significant challenges due to weak geological formations that threaten wellbore integrity. Cementing is a critical process in addressing these challenges and foamed cement has emerged as a viable low-density solution. Its lightweight composition helps maintain balanced hydrostatic pressure, preventing fracturing and ensuring effective zonal isolation and structural support. Advanced technologies optimize cementing operations ensuring the delivery of safe and durable wells.

Foamed cement is characterized by its stable and lightweight composition which provides effective zonal isolation and structural support. Design considerations include precise calculations of foam quality, density, and volume to maximize its efficacy. Practical experiences have demonstrated its practical application though operational challenges such as issues with DV tools can affect cementing reliability.

The paper presents an example that highlights the successful application of foamed cement in a candidate well offering valuable insights into best practices from design to implementation. Test results demonstrated the effectiveness of foamed cement under the typical conditions underscoring the need for comprehensive planning and monitoring to address operational challenges. Through the detailed example, the paper provides practical guidance on foamed cement application serving as a detailed resource for engineers and operators. It covers the properties and benefits of foamed cement along with strategies for operational application, and offers solutions to common operational issues. Therefore, making it an essential guide for optimizing cementing operations in challenging environments.

Foam Cement Definition

Foamed cement is a specialized form of lightweight cement. It has gained attention in the oil and gas industry due to its unique properties and benefits. Foamed cement enhances flow characteristics and offers structural support in challenging formations. It does this by incorporating microscopic gas bubbles, typically nitrogen, into the cement slurry, as shown in [Figure 1](#). This technique solves the problem of formations that cannot handle high hydrostatic

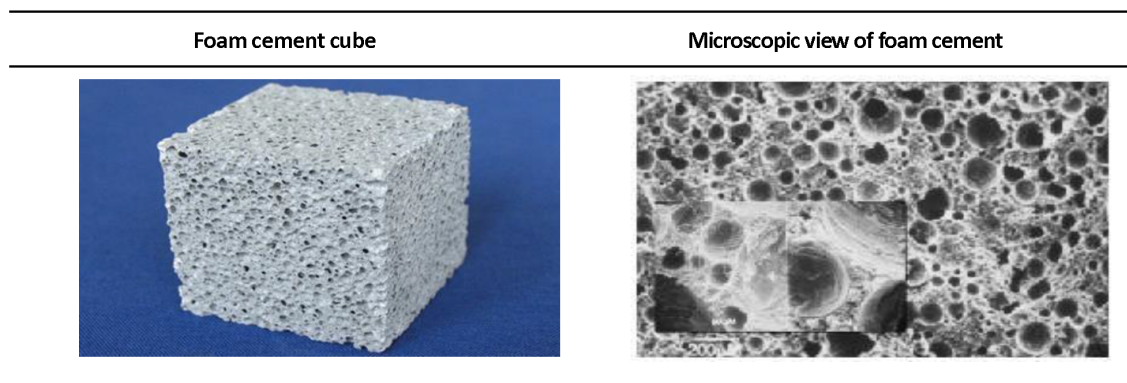


Figure 1—Microscopic View of Foam Cement Bubbles

pressures from traditional cement slurries.

Understanding the Composition and Process

The creation of foamed cement begins at the surface where the cement slurry is prepared. The process involves injecting foaming agents and stabilizers into the slurry followed by the introduction of high-pressure nitrogen to create foam. This setup is common in the oil and gas industry where precise control over the cement's density is crucial for successful well operations. The ability to adjust the density by varying the gas content at the rig site allows operators to tailor the cement properties to specific well conditions without significantly altering other slurry characteristics, making foamed cement a versatile solution. Once the foamed cement is generated, it is pumped into wells through high-pressure lines. It is designed to cure under specific downhole conditions transitioning from a liquid to a solid state. During this curing phase, factors such as temperature, pressure, and shear rate play crucial roles in determining the size and stability of the nitrogen bubbles within the cement. These bubbles, once trapped in the solid matrix significantly influence the mechanical properties of the cement such as its compressive strength and durability. A typical set up for foam generation is shown in [Figure 2](#).

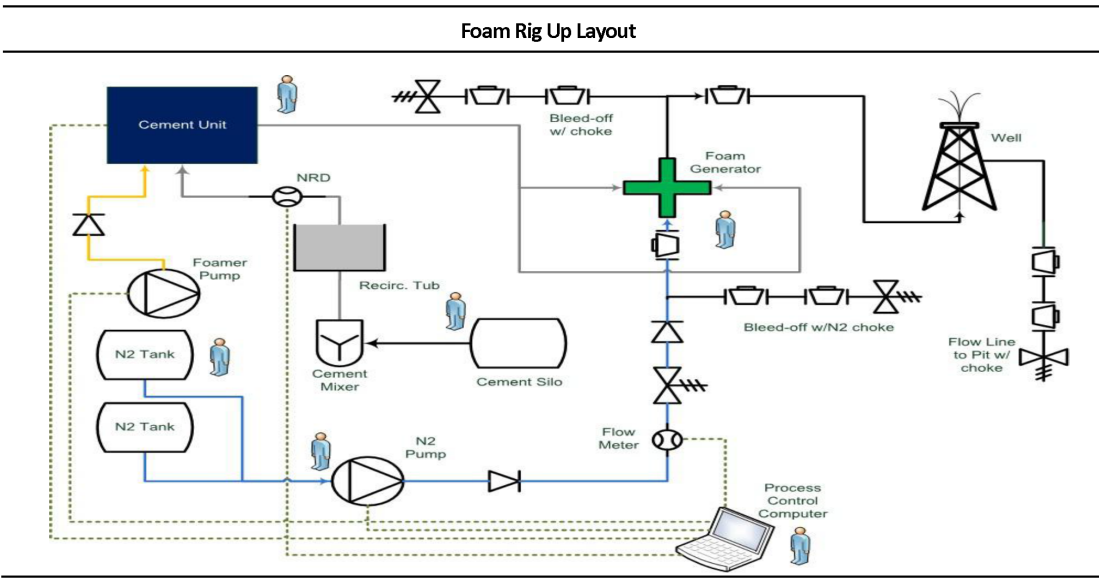


Figure 2—Equipment Layout

Classification by Foam Quality

Foamed cements are classified based on foam quality which is defined by the percentage of gas within the cement volume. This classification is vital as it determines the performance and stability of the cement once cured. A stable foamed cement exhibits a uniform distribution of spherical bubbles which prevents gas breakout and ensures consistent density and strength. This uniformity is crucial for maintaining the integrity of the cement and preventing issues such as channeling which can lead to inconsistent density and reduced compressive strength. Conversely, unstable foamed cements may have irregular voids and bubble distributions resulting in channeling and increased permeability. These characteristics can compromise the structural integrity of the cement leading to potential failures in well operations. Industry practices document such API RP 10B-4, which provide guidelines for cement testing does not fully account for changes in properties during placement highlighting the need for careful monitoring and control during the application of foamed cement.

Table 1

Foam Quality / Ratio is characterized by the ratio of Nitrogen gas volume to the foam slurry volume expressed in percentage	
$F_Q = \frac{V_{N2}}{V_{N2} + V_{BS}} * 100 = \frac{V_{N2}}{V_{FS}} * 100 \dots \dots \dots (1)$	V: Volume N2: Nitrogen BS: Base slurry FS: Foam slurry
Foamed Cement Is Characterized By FQ And Density Of Nitrogen And The Base Slurry	
$\rho_{FS} = \frac{F_Q}{100} * \rho_{N2} + \frac{100 - F_Q}{100} * \rho_{BS} \dots \dots \dots (2.a)$	ρ_{FS} : Downhole Foam Density ρ_{N2} : Downhole Nitrogen Density ρ_{BS} : Density Of The Base Slurry
Or $\rho_{FS} = \rho_{BS} - \frac{F_Q}{100} * (\rho_{BS} - \rho_{N2}) \dots \dots \dots (2.b)$	
Or $F_Q = \frac{\rho_{BS} - \rho_{FS}}{\rho_{BS} - \rho_{N2}} * 100 \dots \dots \dots (2.c)$	
Foam Yield Has The Same Definition As Slurry Yield But With Added Nitrogen Volume	
$Y_{FS} = \frac{Y_{BS}}{100 - F_Q} * 100 \dots \dots \dots (3)$	Y_{FS} : Yield Foam Slurry Y_{BS} : Yield Base Slurry

Foam cement advantage

One of the primary advantages of foamed cement is its low density which makes it suitable for a wide range of applications particularly in challenging geological conditions. Experiments have demonstrated that foamed cement can achieve low densities allowing for its use in formations where traditional cement might be too dense and exert too much pressure. This low-density characteristic is particularly beneficial in the oil and gas industry where maintaining wellbore stability and preventing formation damage are critical. Nitrogen, the gas typically used in foamed cement is considered inert because it does not affect the cement hydration reaction. This inert nature ensures that the addition of nitrogen does not interfere with the cement's ability to set and harden maintaining the integrity of the wellbore structure. Additionally, foamed cement possesses anti shrinkage properties which inhibit fluid migration and reduce damage to water-sensitive formations. These properties make foamed cement particularly useful in zones that experience total losses where traditional cement might fail to provide adequate support.

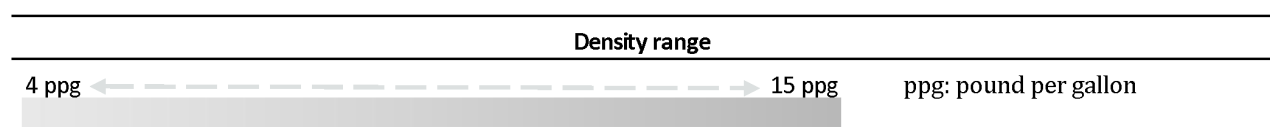


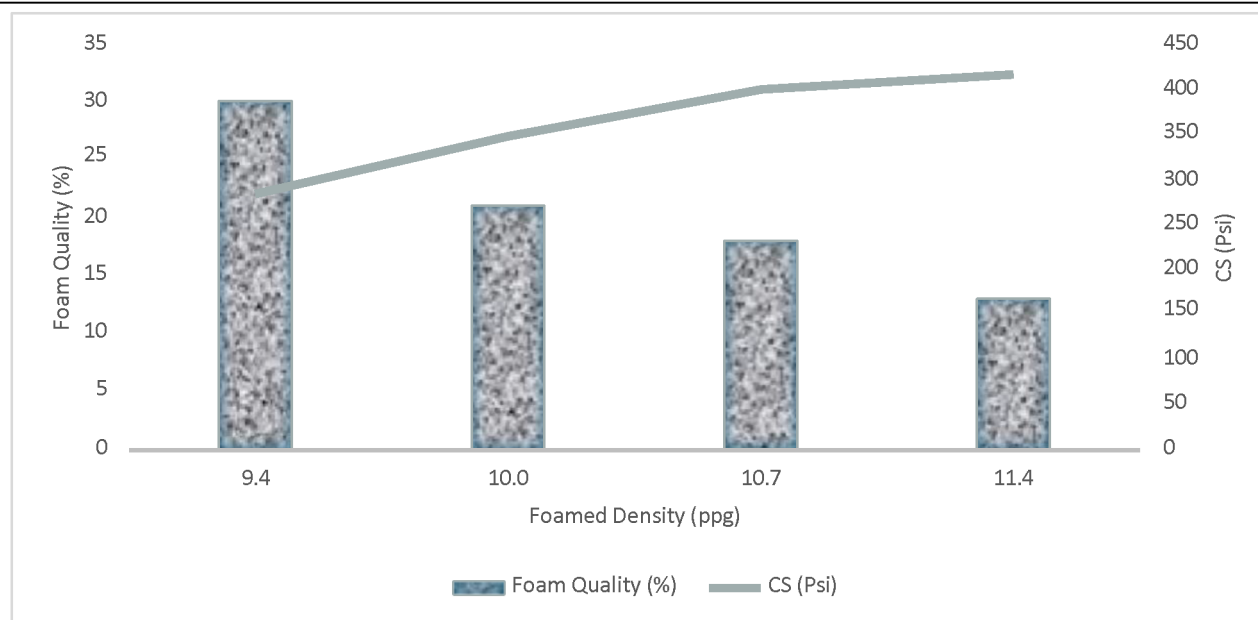
Figure 3

Mechanical Properties

Foamed cement is characterized by high durability and flexibility offering excellent mechanical properties that make it a preferred choice in various applications. Its ability to offer structural support while maintaining a low density which helps prevent damage to fragile geological formations by minimizing hydrostatic pressure as illustrated in [Figure 4](#). The mechanical properties of foamed cement such as its compressive strength and durability are significantly influenced by the nitrogen bubbles trapped within the solid matrix. Its versatile nature allows for adaptation to diverse well conditions making it effective in both vertical and horizontal well designs. By tailoring cement properties to specific needs such as adjusting density or adding performance-enhancing additives. Foamed cement provides reliable stability and protection, making it an ideal choice for complex and challenging well environments.

Foam Cement Mechanical Properties within 24 hours

Base Density ppg	BHCT degF	H ₂ O/Cement Ratio (%)	Micro silica (BWOC)	Viscosifier (BWOC)	Retarder (BWOC)	Surfactant (Gal /Sk)	Foamer (Gal /Sk)
12.7	155	26.2	15	0.3	0.650	0.196	0.196

**Figure 4—Foam Cement Mechanical Properties**

Foam Cement Design

The design process involves two primary components: laboratory design and pre-job planning. Laboratory design evaluates the properties of foamed cement including compressive strength and stability under various conditions with the aim of adjusting slurry properties for optimal placement and long-term performance. Pre-job planning and engineering consider factors such as pump rates, pressures and geological conditions of the well to develop a strategy that ensures successful cement placement.

Foamed Cement Laboratory Test

Initially, a base cement slurry was prepared and its properties were measured in the un-foamed state. This base slurry was then foamed after which its stability was assessed and analyzed across various slurry densities at the required foam quality

The design of foamed cement treatments consists of two key parts:

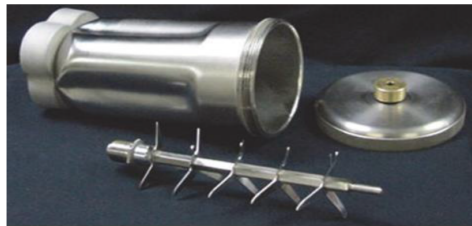
1. Laboratory design to adjust the slurry properties to ensure the cement is correctly placed and has the desired performance over its lifetime.
2. Performance examination of the foamed cement slurry in the laboratory following recommended practice of API RP 10B.

Test Procedure Overview

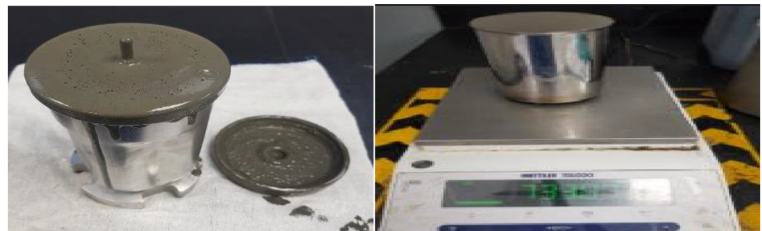
Table 2

Test Step	Description	Equipment/Standards Used	Notes
Overview	Overview of the foamed cement slurry laboratory test procedure.	API RP 10B-4, ISO 10426-4	The design of foamed cement treatments consists of laboratory design to adjust slurry properties.
Preparation of Base Slurry	Preparation of the base cement slurry and measurement of its properties in the un-foamed state.	Blending container with a lid, mixing blade assembly	The blending container should have a threaded cap with an O-ring seal.
Preparation of Foamed Slurry	Foaming the base slurry and assessing its stability across various slurry densities.	Blending container with a lid, mixing blade assembly	Foam quality is evaluated based on density variation within ± 0.2 ppg of the base slurry.
Determination of Foamed Cement Slurry Density	Checking the atmospheric foamed slurry density to ensure it matches the theoretical density.	Open-top container with known volume and weight	Avoid using pressurized fluid density balance to prevent inaccurate density readings.
Stability (Unset Foamed Slurry)	Evaluating the stability of unset foamed cement slurry using a graduated cylinder.	250-ml graduated cylinder, API 10B-2 Free Fluid Test	Seal the top of the cylinder to prevent dehydration and leave for 2 hours at ambient temperature.
Stability (Set Foamed Cement Slurry)	Evaluating the stability of set foamed cement slurry using a curing mold and density measurement.	Curing mold, water bath at BHCT, API RP 10B-4 section 9.4	Cut the sample into six pieces and measure the density of each piece (top, middle, bottom).
Determination of Compressive Strength	Measuring the compressive strength of the foamed cement slurry after curing in a sealed mold.	Curing mold with sealing lid	The sealing lid prevents the foamed cement slurry from expanding as it is heated.

Equipment



Blending Container and Multi-blade Assembly



Foam Cement densities measurement



Curing Mold: Foamed Cement Slurry Stability



Curing Mold: Compressive Strength

Figure 4a—Laboratory Equipment

Engineering Design

The design methodology for foamed cement jobs focuses on two main objectives: maintaining the cement column's hydrostatic profile within pore and fracture pressures during circulation and achieving a final density profile that provides adequate compressive strength for casing support and zonal isolation. To effectively design the treatment, it's crucial to prioritize simplifying operations to reduce complexity and errors and maximize cement usage while minimizing nitrogen for enhanced process effectiveness. Four key parameters can be adjusted to optimize foamed cement design by the number of stages, the nitrogen ratio for each stage, the backpressure applied on the annulus, and the cap slurry's length and placement method. Adjusting the number of stages helps regulate pressure and achieve the desired density. Altering the nitrogen ratio affects foam quality and cement density while applying appropriate backpressure maintains hydrostatic balance and prevents gas escape. The cap slurry's length and placement method ensure effective gas compression and cement integrity. This comprehensive approach reduces risks that the foamed cement operation fulfills structural and isolation needs while ensuring operational efficiency.

Table 3

Design Principles	Description
Goals	Maintain hydrostatic profile between pore and fracture pressures during circulation. Achieve a density profile that provides compressive strength for casing support and zonal isolation.
Main Ideas	-Simplify the operation as much as possible. -Maximize cement usage in the well, minimizing nitrogen use.
Number of Stages	Determine the appropriate number of stages to manage pressure and achieve desired density. Figure.6
Nitrogen Ratio	Adjust nitrogen ratio for each stage to control foam quality and cement density.
Backpressure	Apply suitable backpressure on the annulus to maintain hydrostatic balance and prevent gas escape.
Cap Slurry Length and Placement	Decide on the length of the cap slurry and the method of placement for effective gas compression and cement integrity.

Generic Well Schematic

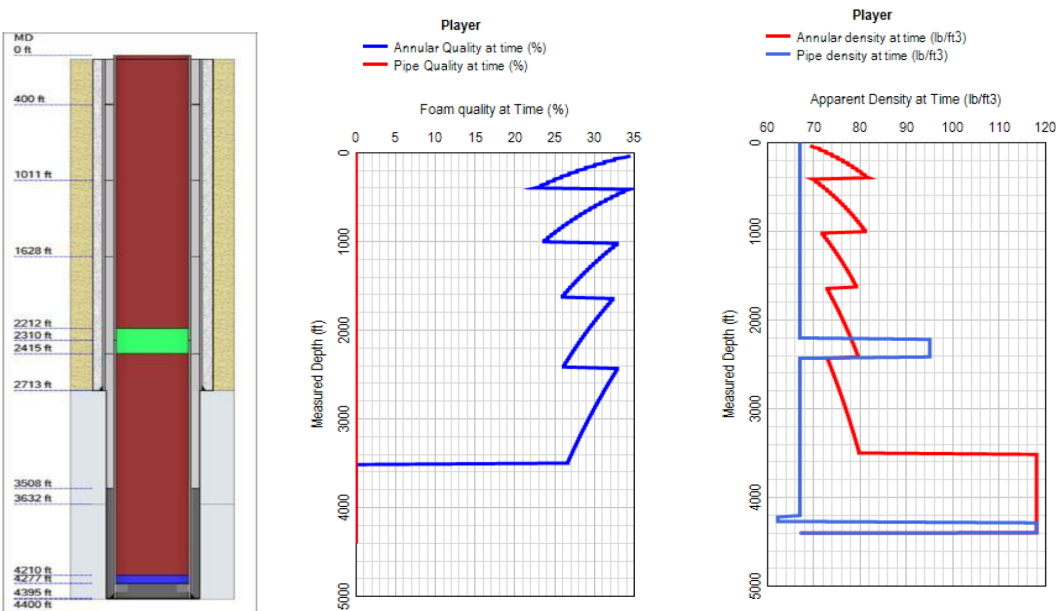


Figure 5

Field Application of The Foam Cement

Geological Aspects

The geological formation in the subject well poses considerable drilling challenges because of its weak characteristics which compromise wellbore integrity through lost circulation and pressure management issues. Lost circulation occurs when drilling fluids escape into porous or fractured formations posing threats to well integrity and operational efficiency. Effective cementing operations are crucial to addressing these challenges especially in areas susceptible to washout, incomplete annulus fill, and casing collapse.

Field Application

Drilling operations in similar wells, specifically the intermediate casing section involves two significant challenges. The first challenge is the water flow formation encountered in shallower formation which can be managed by drilling with controlled water flow using a 73 pcf (9.8 ppg) mud weight until reaching the weak lost circulation zones formation, where complete circulation loss can occur. Common practices in such scenarios include drilling while the water flow formation is flowing or mud capping the well and continuing with drilling operations. Cementing the casing typically occurs in two stages, the first stage fills the annulus up to the lowest lost circulation zone, followed by a second stage using a DV stage tool set inside the previous casing to shut off the water flow. However, this approach can leave the casing exposed to corrosion. The second challenge is the risk of lost circulation which heightens the risk of complications during cementing operations. Multistage cementing often faces issues such as mechanical failures, cement channeling, and high costs which are diminishing the success rates of primary cementing jobs. The fragile nature of the formations in this area demands specialized techniques to ensure well stability and integrity. Utilizing foam cement can enhance cementing reliability and efficiency, improve well integrity, and mitigate the risks associated with drilling in these challenging geological formations.

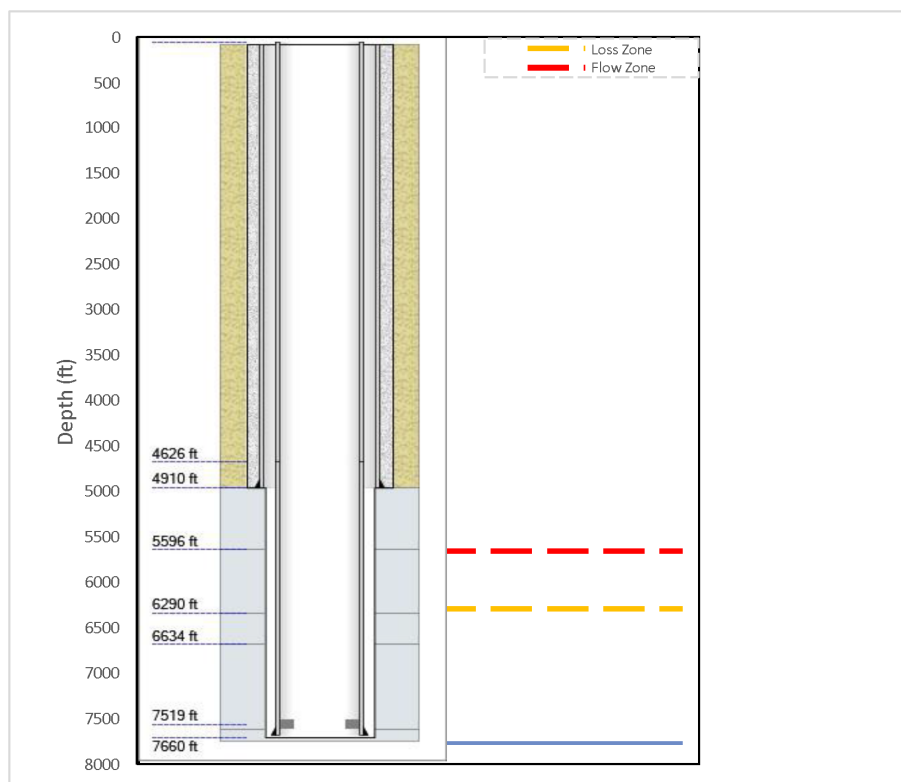


Figure 6—Well Diagram

Design methodology

Various best practices were implemented to mitigate risks encountered when drilling similar wells. The goal was to maintain hydrostatic pressure while drilling the 12-inch open hole section and setting the intermediate casing with a density of 73 pcf (9.8 ppg). The original plan was to cement the casing in two stages using a DV tool. However, the primary strategy was to use foamed cement to bring it to the surface in a single stage. The DV tool was kept ready with the capability to be canceled if not required. If the foamed cement did not reach the surface as planned, the DV tool would be activated to proceed with a second stage cementing operation.

Operation Summary

- **Circulation and Hole Condition:** After running the casing to the bottom, the rig circulated two bottoms-up at 11-12 bpm and around 350 psi, with stable pressures and no losses. This indicated that the wellbore was stable. Pump a marker pill down to check the wellbore condition, which revealed an under-gauged open hole upon recovery.

Cementing Job Design

- **Slurries:** The cementing job was designed with two types of slurries:
 - **Tail Slurry:** Conventional cement with a density of 15.8 ppg.
 - **Lead Slurry:** Foamed cement with a base density of 12.7 ppg, foamed over six nitrogen stages to achieve downhole densities ranging from 9.0 ppg to 10.2 ppg. This corresponds to a foam quality range of 30% to 20%.
- **Marker Pill:** lost circulation fibers were added to the first 50 bbl of the lead slurry as a marker pill.
- **Spacer:** A 10.7 ppg pre-flush spacer was used, also foamed with 30 scf/bbl to aid in efficient displacement and ensure proper cement placement.
- **Cap slurry:** pumped directly down in the annulus following the top plug bumping to compensate for the backpressure.

Best Practices Applied

- **Hydrostatic management:** Throughout the operation, maintaining hydrostatic pressure was prioritized to prevent formation fluid influx or losses.
- **Foamed Cement:** The use of foamed cement helps reduce hydrostatic pressure and improve zonal isolation.
- **DV Tool Strategy:** The stage tool was kept with cancellation capabilities, providing flexibility to switch to a two-stage cementing operation if needed.

This approach highlights the importance of contingency planning in well cementing operations, especially in a challenging environment. By preparing for potential issues with foamed cement reaching the surface, the team ensured they could adapt quickly and maintain well integrity.

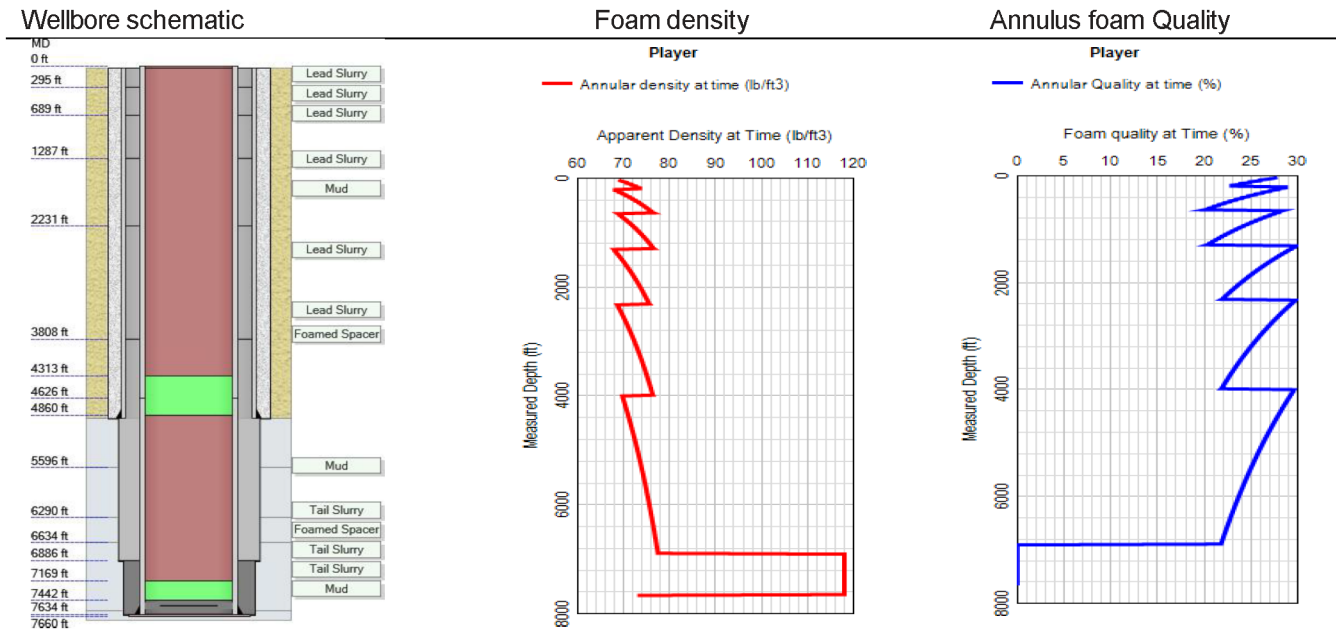


Figure 7

Fluids Descriptions

Table 4—Lead proprieties

Lead Foamed Slurry							
Base Density	Temp	H ₂ O/Cement Ratio	Micro silica	Viscosifier	Retarder	Surfactant	Foamer
ppg	degF	(%)	(BWOC)	(BWOC)	(BWOC)	(Gal /Sk)	(Gal /Sk)
12.7	155	26.2	15	0.3	0.650	0.196	0.196

Table 5—Tail proprieties

Tail Slurry						
Density	Temp	H ₂ O/Cement Ratio	Antifoam	Anti-Settling	Dispersant	Retarder
ppg	degF	(%)	(Gal /Sk)	(BWOC)	(Gal /Sk)	(BWOC)
15.8	155	41.26	0.002	0.20	0.20	0.25

Table 6—Spacer proprieties

Spacer				
Base Density	Temp	Mix water	Viscosifier	Weighting agent
ppg	degF	(gal/bbl)	(lb/bbl)	(lb/bbl)
10.7	155	36.98	5.500	127.341

Table 7—Foam Cement Laboratory testing

Atmospheric Testing			
BHCT (°F): 155 degF			
Lab Test Results	Original Design	- 0.2 ppg	+ 0.2 ppg
Base Slurry Density (ppg)	12.70	12.50	12.90
Avg Foam Density (ppg)			
Designed Quality (%)	30%	30%	30%
Section 7.2 of API RP 10B-4			
Fill Cup After 15 sec?	NO	NO	NO
If no, time to fill (sec)	4	4	5
Final RPM	8128	8072	8106
Section 8.3 of API RP 10B-4			
Required downhole Foam density (ppg)	8.93	8.73	9.13
Calculated Foam density at atmospheric (ppg)	8.89	8.76	9.04
Measured Foam density at with offset correction (ppg)	8.87	8.81	9.10
Section 8.4 of API RP 10B-4			
Free Fluid (ml)	0	0	0
% Reduction	<1%	<1%	<1%
Time for Unset Foam Density (hours)	2	2	2
Top Density (ppg)	8.85	8.73	9.05
Middle Density (ppg)	8.88	8.78	9.08
Bottom Density (ppg)	8.97	8.85	9.17
Time for Set Foam Density (hours)	24	24	24
1. Top Density (ppg)	8.85	8.73	9.07
2. Next sample (ppg)	8.91	8.75	9.12
3. Next sample (ppg)	8.93	8.81	9.15
4. Next sample (ppg)	9.01	8.70	9.19
5. Next sample (ppg)	9.03	8.88	9.22
6. Bottom Density (ppg)	9.08	8.93	9.23
Photo Before Cutting			
Photo After Cutting			

Job execution and evaluation

The foam cementing process begins with preparing a base cement slurry, conducting safety checks and calibrating equipment to ensure safe operations. Nitrogen and a foaming agent are added to the slurry to create a stable foam structure that reduces its density. This foamed slurry is then pumped downhole with careful management of pressure and flow rates adhering to a strategic plan for optimal cement placement. Continuous pumping is maintained until foam returns are observed at the surface confirming successful placement.

Throughout the foam cementing job, full returns were maintained which indicates stable well conditions. Foam returns were visible at the surface with annulus pressure around 350 psi and the foamed spacer and cement were observed. No foam expansion occurred in the waste pit eliminating the need for a defoaming solution. The top plug was bumped to 1600 psi, and choke pressure of 300 psi was applied to stop persistent foam cement returns after the job. A cap slurry was pumped in stages to equalize annular pressure at approximately 300 psi. Once the final cap slurry volume was bullheaded and the Side Outlet valve (SOV) was opened, foamed returns ceased allowing operations to continue smoothly. This included dropping the DV cancellation plug to close the DV tool and proceeding with wellhead work ensuring a uniform transition from cementing to subsequent well operations.

Job Pumping Charts

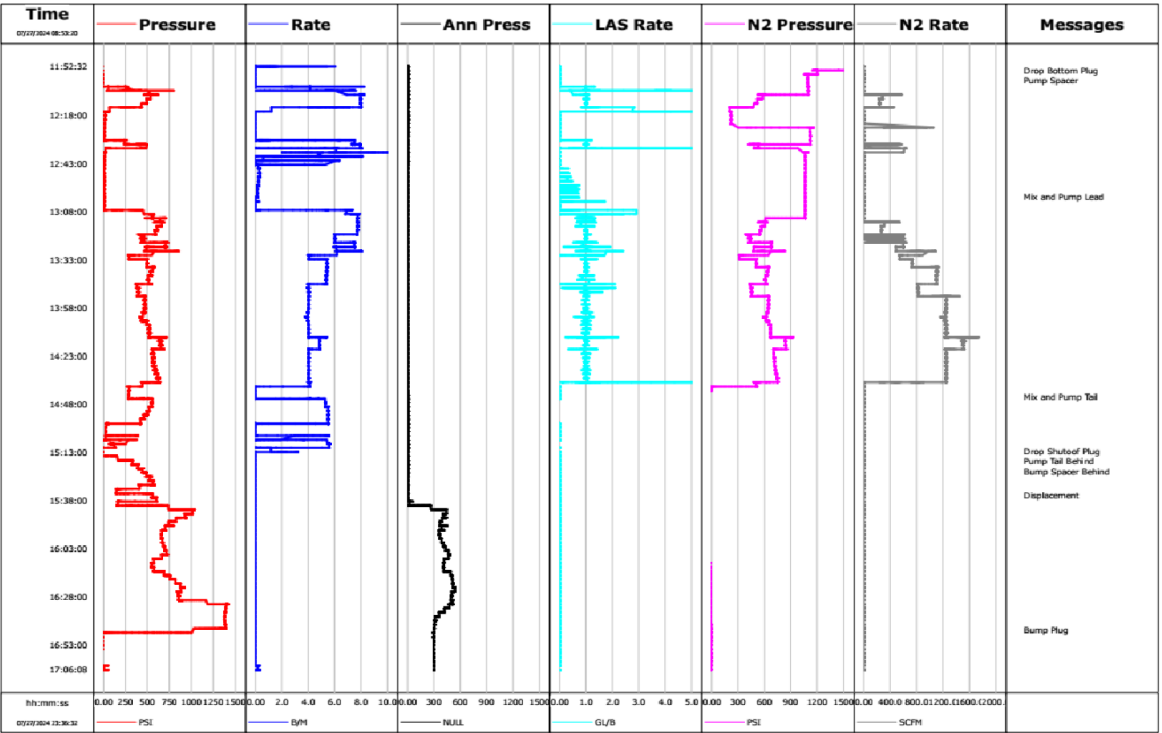


Figure 8—Pumping Charts

The pressure match validated the cementing execution confirming there were no losses and showing clear lift of cement in the annulus. The effective job placement was evidenced by the correlation between actual and simulated pressures in Pressure Match Figure 9, demonstrating reliable execution. The lead slurry maintained an average density of 9.76 ppg, while the applied back pressure and a strategic pumping cap slurry schedule effectively managed return. Despite challenging downhole conditions involving varying temperature, pressure, and nitrogen expansion the process was well-coordinated and executed smoothly. Highlighting the efficiency and precision of the cementing operation.

Pressure Match

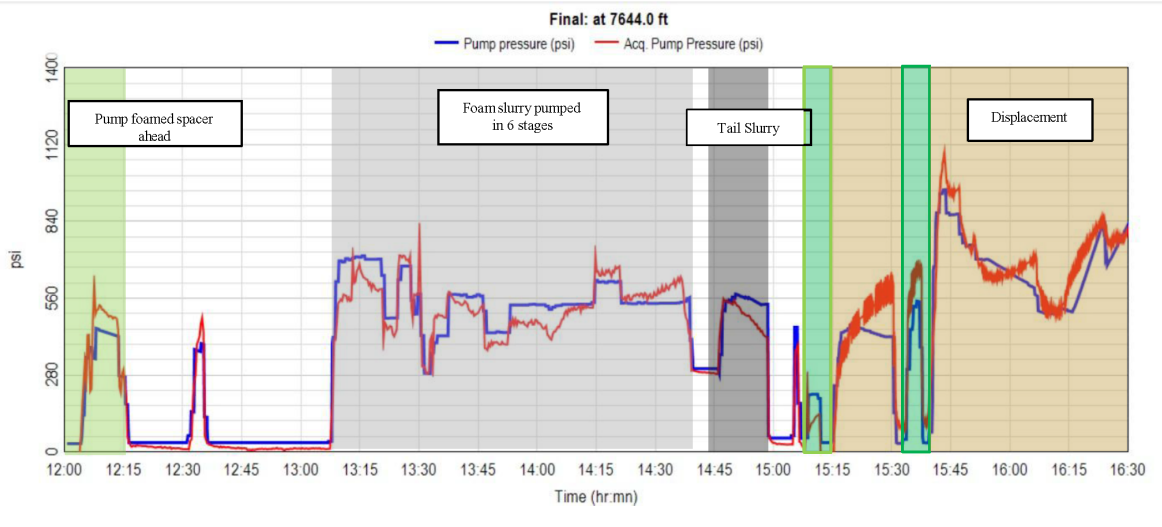
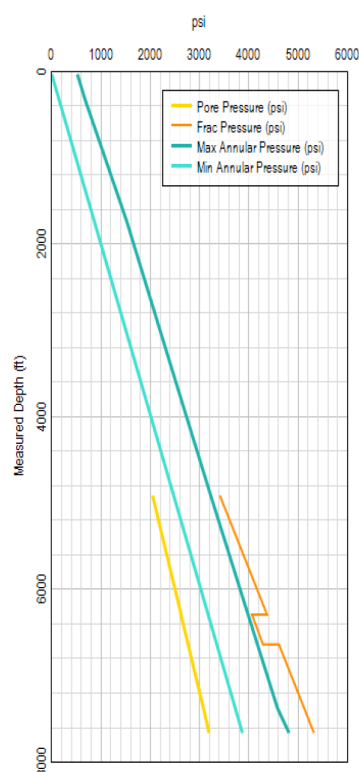
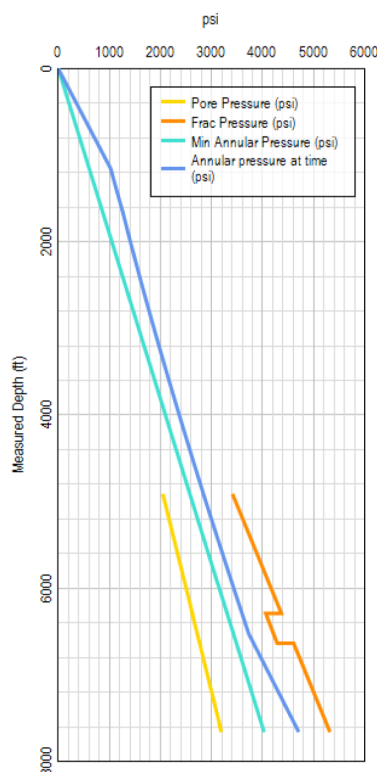


Figure 9—Pressure Match

Well Security During the Cement job



Well Security After Cap Slurry



Annulus foam Quality

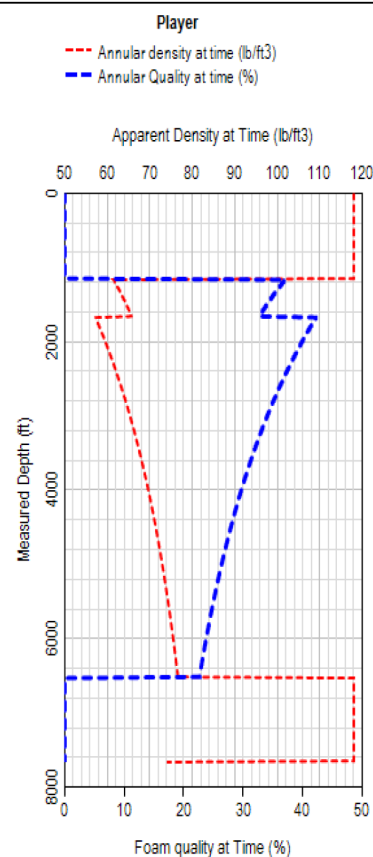


Figure 10—Well Security

Conclusions and Recommendations

The deployment of foamed cement has proven to be a highly effective solution for addressing the challenges posed by geologically weak formations particularly in zones susceptible to washout, lost circulation, and casing collapse. This innovative cementing method mitigates the risks associated with DV stage tool failures by ensuring consistent foam quality and uniform slurry distribution thereby enhancing wellbore integrity.

Key conclusions include

- **Improved Well Integrity:** Foamed cement provides superior zonal isolation and structural support, reducing the likelihood of formation damage and fluid migration.
- **Operational Flexibility:** The system allows for contingency planning, such as switching from single-stage to two-stage cementing if needed, without compromising job success.
- **Cost and Risk Reduction:** By minimizing the need for remedial operations and reducing the risk of mechanical failures, foamed cement contributes to more efficient and economical well construction.

Recommendations for Future Applications

1. **Returned Volume Analysis:** Use wireline caliper logs or marker pills to verify borehole size before casing installation.
2. **Flow Area Optimization:** Distribute Side-Outlet Valves (SOVs) evenly around the casing to enhance return flow and ensure even cap slurry placement.

3. **Open Hole Excess (OH Excess):** Begin with a conservative estimate (0–20%) and adjust based on marker pill results and pre-job circulation loss data.
4. **Placement Analysis:** Conduct excess sensitivity analysis to optimize slurry volumes and ensure the second foam stage remains in the well, minimizing the need for backpressure or cap slurry adjustments.

This approach, initially applied in onshore wells, is also suitable for offshore and complex environments including shallow hazards, depleted zones and unconsolidated formations. Foamed cement stands out as a robust, adaptable and cost-effective solution for wellbore integrity challenges.

Nomenclature

BHST:	Bottomhole Static Temperature
BHCT:	Bottomhole Circulating Temperature
ECD:	Equivalent Circulating Density
BWOC:	By Weight of Cement
GPS:	Gallon Per Sack
OH:	Open Hole

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